



Introduction to the Microscope

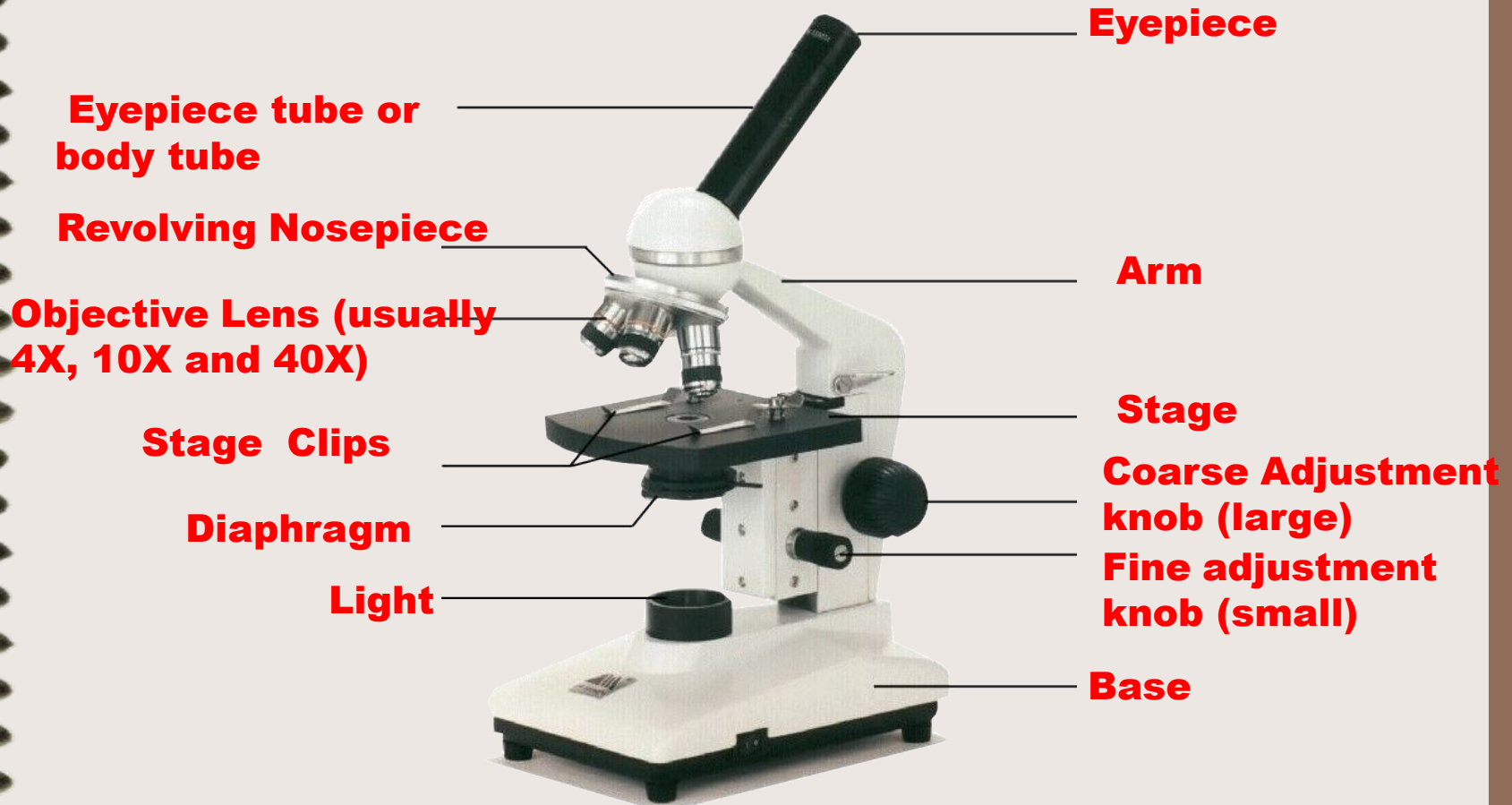
- Care
- Parts
- Focusing

Microscope Care

- Always carry with 2 hands
- Only use lens paper for cleaning
- Do not force knobs
- Keep objects clear of desk and cords

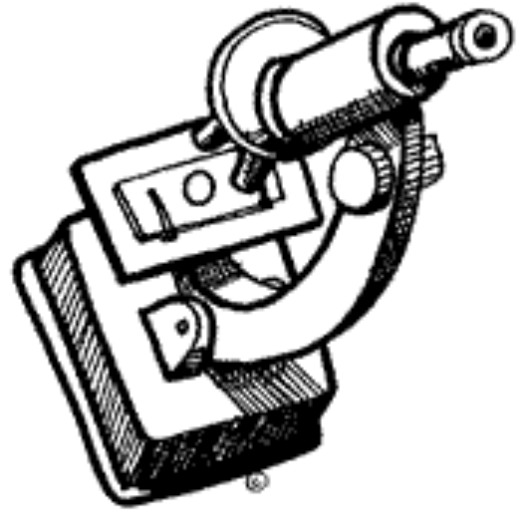


Microscope Parts



Using the Microscope

- Place the Slide on the Microscope **stage**
- Use **Stage Clips** or arm
- Always start on the **lowest objective**
- Look into the **Eyepiece**, you will see a **pointer (arrow)**
- Use the **Coarse Focus FIRST**
- Use the **Fine Focus** to make the image clear.

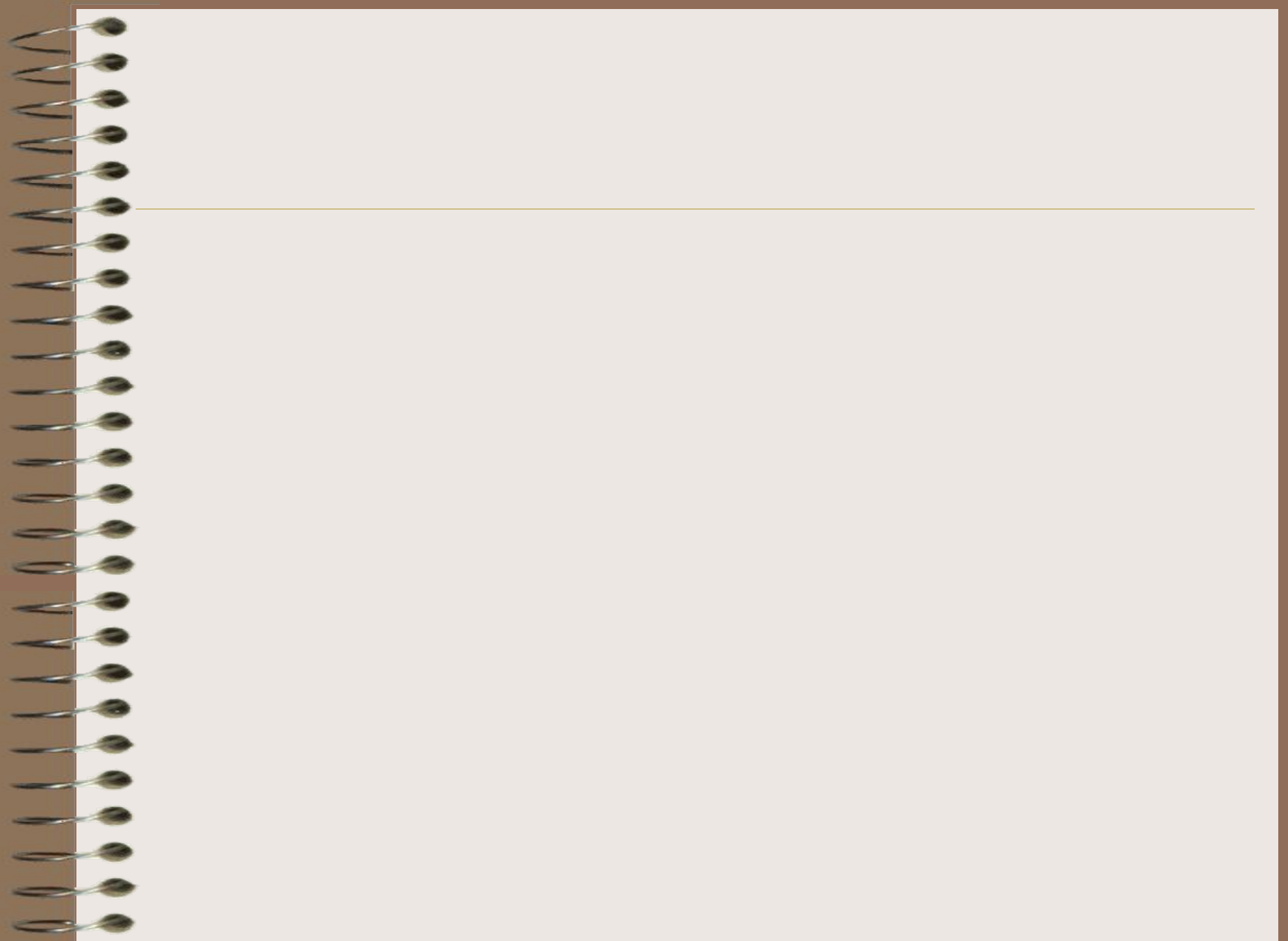


Using High Power

- Click the objective to the next highest power, get the image in **focus**, you should here the objective **click into place**
- When you get to the **largest objective**
- **Do NOT** use the **Coarse Focusing Knob**.
Use the Fine Focus Knob **ONLY**



What can you find on your slide?



Tools used in Science...

- **Microscopes**

- **Magnification** – determined by multiplying the eyepiece times the objective
- e.g. If you are looking into the **eyepiece** (usually **10X**) using the **shortest objective** (usually **4X and red**), total magnification is $10X * 4X = 40X$ **magnification**
- **Resolution** – ability to distinguish between two objects

Compound Light Microscope



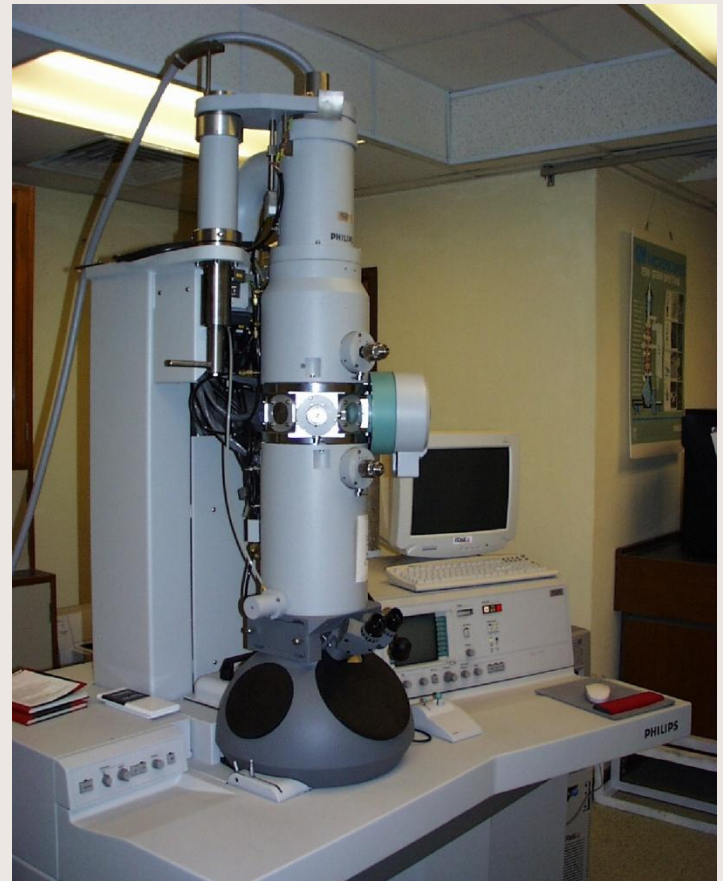
- Light passes through the object
- Can view living specimens
- Not a lot of detail

Compound Light Microscope

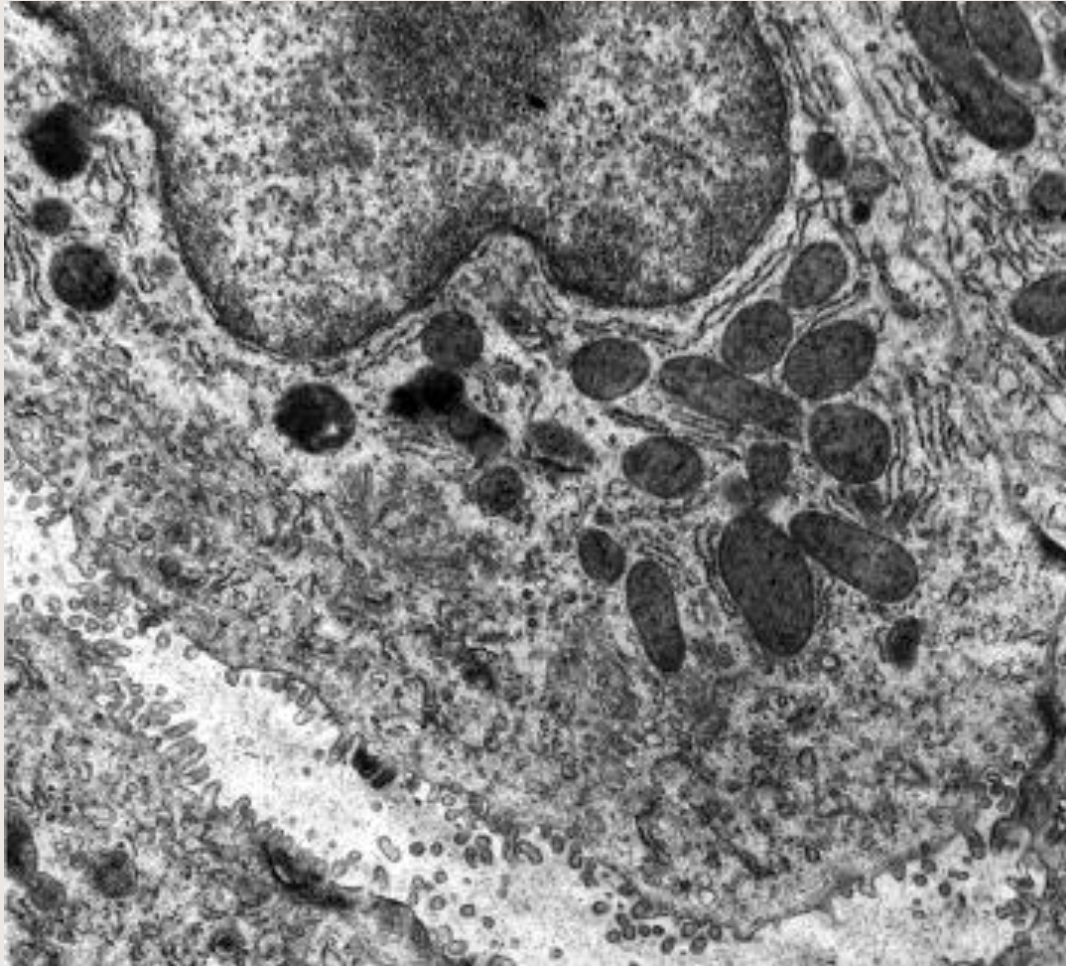


Transmission Electron Microscope (TEM)

- Passes a beam of electrons through the object
- Only dead cells can be viewed
- Cells sliced thin
- Much greater detail



Transmission Electron Microscope



Scanning Electron Microscope (SEM)



- Beam of electrons sweeps over and bounces off of object
- Produces a 3-D image
- Less magnification than TEM

Scanning Electron Microscope

